

Linkset Usage Pattern: Content Negotiation Menu

Pattern Name

Content Negotiation Menu

Goal

The objective of this design pattern is to resolve the mostly “hidden nature” of content-negotiation implementations on the web.

To complement the existing practice this pattern encourages such implementations to expose the available variants and its possible interplay with redirection-practices.

Motivation

Half-sided “negotiation”

In standard HTTP content-negotiation there is actually only a very limited level of negotiation-dialogue going on. Compared to the food-ordering conversation in a restaurant which starts with asking and browsing the menu of what is available, or even be guided to suggestions by a helpful waiter, the web model more reflects the reality of ordering a drink at a noisy high throughput festival-bar: you get to shout your preferred order, but will have to content yourself with whatever you get back.

Indeed, being confined to only one request-response cycle, the client can express its preference in HTTP-Accept headers, but still has to simply deal with how the server justifies its best offer to satisfy that. Through this linkset-pattern the service (and its both-sides negotiability) can be augmented. By slipping in a linkset that encodes the menu-of-availability in the response-header a client can actively decide whether an alternative additional request promises to better fit its scenario.

The “Broken Chain” Problem

The real world implementations of content-negotiation often rely on the 303 See Other status code to redirect a client from a Conceptual URI (the abstract identity of a resource) to a Representation URI (a specific file format like .ttl, .jsonld, or .html).

This mechanism creates a “Broken Chain” phenomenon for machine agents. Upon following the redirect, the agent arrives at a specific representation but often loses the state and context of the original Conceptual URI. Without explicit protocol-level signaling, the agent becomes “stranded” on a single variant,

losing visibility of the “menu” of other available representations and the semantic metadata associated with the conceptual identity.

The Radical Transparency (RT) framework solves this by enforcing an Identity Anchor within the representation layer. By utilizing rel=“self” for identity persistence. With this the server ensures that the chain of identity remains unbroken and that the full “representation menu” remains discoverable post-redirect.

Applicability

This pattern SHOULD be applied to any resource within an interoperable data space that:

- Exposes a Conceptual URI representing a distinct entity or digital asset.
- Provides multiple representations (variants) through HTTP content negotiation.

Encoding

To ensure machine-actionable transparency, implementations MUST adhere to a number of exposed link-relations at both the conceptual and representation levels.

Additionally, this pattern strongly suggests to explicitly introduce the variant menu level (the linkset) This leads to benefits for both design (overview and manageability) and operational (materialisation and cacheability) aspects of this pattern.

Conceptual Resource Level (Identity)

When a client accesses a Conceptual URI (i.e. one supporting content-negotiation), the server MUST signal the availability of the alternative representations during the response or redirection process.

```
# from the <concept-uri> anchor
Link: <variant1-uri>
      ; rel=alternate; type={mime1}; language={lang1}; profile={prof1},
      <variant2-uri>
      ; rel=alternate; type={mime2}; language={lang2}; profile={prof2},
      ...
      <variantN-uri>
      ; rel=alternate; type={mimeN}; language={langN}; profile={profN},
```

Note: The applied parameters for the effective negotiation (by mimetype, language or profile) are to be independently provided in the link-relation

Representation Variant Level (Content)

Every representation variant (regardless of it being arrived at through any, one, or multiple redirects) MUST include the HTTP Link headers that anchor the variant back to the resource graph:

```
# from any actual <representation-uri> anchor
Link: <concept-uri>; rel=self
```

Variant Menu Level (Options)

The above link-relations can reuseably be coded into a central linkset that functions as a local navigation map between all these resources. This linkset then actually materializes this “menu of variants”.

In that case all uri playing a role in the pattern (i.e. the concept-resource, all representation-variants and the variant-menu) should simply refer to that available variant-menu-linkset through

```
# from any <uri> anchor in this pattern
Link: <variant-menu-lsjson-uri>
      ; rel=linkset; type="application/linkset+json",
      <variant-menu-lstext-uri>
      ; rel=linkset; type="application/linkset",
```

And additionally have it map out the relative roles and relations:

in application/linkset+json syntax:

```
{ "linkset":
  [
    { "anchor"   : "<concept-uri>",
      "alternate": [
        { "href"   : "<alternate-variant1-uri>",
          "type"    : "{mime1}",
          "language": "{lang1}",
          "profile" : "{prof1}",
        },
        ... other available alternate variants described and listed here
      ]
    },
    { "anchor"   : "<alternate-variant1-uri>",
      "self"     : [ { "href": "<concept-uri>" } ]
    },
    .... other alternat-variant-uri listed and linked to rel=self here
  ]
}
```

in application/linkset syntax:

```

<concept-uri>
  ; rel="self"
  ; anchor="<alternate-variant1-uri>,"
<alternate-variant1-uri>
  ; rel="alternate"
  ; type="{mime1}"
  ; language="{lang1}"
  ; profile="{prof1}"
  ; anchor="<concept-uri>",

... above repeated for N variants

```

Note on search index optimisation and rel=canonical

A close relative to the **rel=self** relation is the **rel=canonical**, introduced in RFC 6596 to mark the “preferred URI” to use for a specific resource:

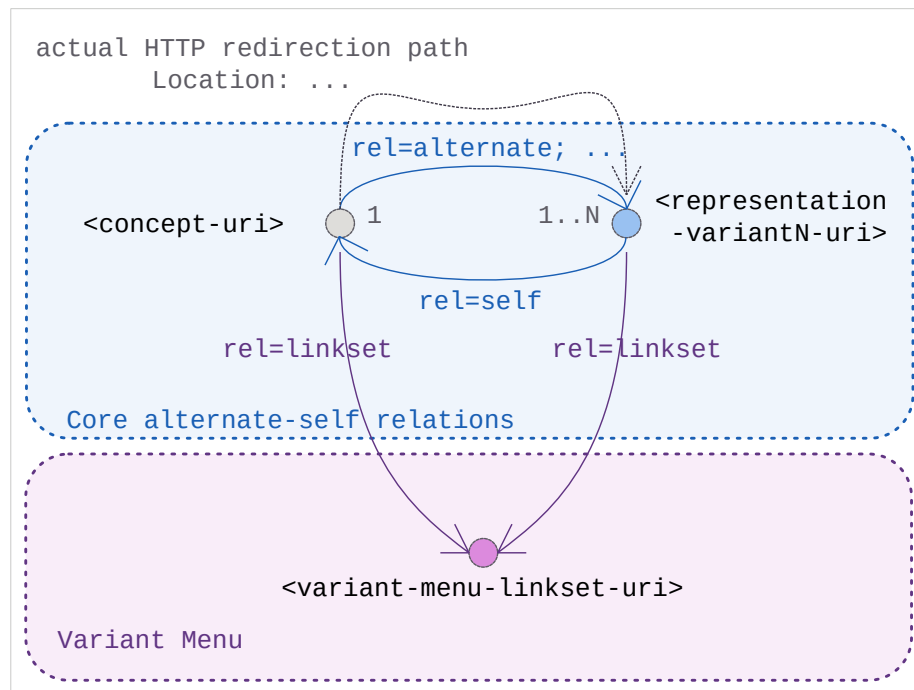
In regard to the link relation type, “canonical” can be described informally as the author’s preferred version of a resource. More formally, the canonical link relation specifies the preferred IRI from a set of resources that return the context IRI’s content in duplicated form. Once specified, applications such as search engines can focus processing on the canonical, and references to the context (referring) IRI can be updated to reference the target (canonical) IRI.

The prime application domain targetted here has clearly been search-engines. Given the textual references and the list of cited implementations in the appendix A of the rfc this search-engine application-domain is likely even the sole trigger for this: avoiding some canabilisation of “search-index-ranking” caused by the avoidable competition introduced by variants of essentially the same content from the same source.

It should be clear the chosen **rel=self** label in this pattern is distinct from this, targetting the ‘identifier’ role, the one resource that caters for content-negotiation, not simply the ‘preferred’ one.

Still, in cases where search-engine references is important for your resources, adding the **rel =canonical** to the pattern should surely be considered. In such cases it should typically be (i) either duplicating the resource marked as **rel=self** or (ii) aiming for the human-focus of search-engine use it to directly mark the human-oriented variant (typically the **type=text/html**).

Sketch



Sketch of the linkset-usage-pattern for content-negotiation menu's

Link Relations Used

The following link relations are in use to describe the links between a core conneg supporting identity and the alternatives it offers:

Relation Type	Specification Source	Technical Function
rel="self"	RFC 4287 - Atom Syndication Format	Acts as the Identity Anchor at each negotiation endpoint; it links the specific variants back to the central conceptual URI to restore the "Broken Chain" after redirects

Relation Type	Specification Source	Technical Function
rel="alternate"	RFC 8288 - Web Linking	Signals the availability of alternative representation variants (such as Turtle, JSON-LD, or HTML) for the conceptual resource
rel="linkset"	RFC 9264 - Linkset	Refers to the external Variant Menu (the linkset) in which all available options and their interrelationships are summarized in a machine-readable format.
rel="canonical"	RFC 6596 - The Canonical Link Relation	Refers to the authors' «preferred» variant. The one that is typically returned by search-engines.

See IANA Link relations

Implementation Example: MarineInfo Case Study

In this scenario, a machine agent requests the identity resource for a specific marine observation.

Conceptual URI: <https://marineinfo.org/id/institute/36>

Target Variant: <https://marineinfo.org/id/institute/36.html>

Target Variant: <https://marineinfo.org/id/institute/36.ttl>

Target Variant: <https://marineinfo.org/id/institute/36.jsonld>

Variant Menu: <https://marineinfo.org/id/institute/36-ls.json>

Conneg Request for text/turtle

The client requests the conceptual resource with a preference for turtle. The server redirects the agent while providing the linkset context.

```
curl -I "https://marineinfo.org/id/institute/36" -H "Accept: text/turtle"
```

HTTP/1.1 303 See Other

Location: <https://marineinfo.org/id/institute/36.ttl>

Link: <<https://marineinfo.org/id/institute/36-ls.json>>; rel="linkset"; type="application/ld+json"

Representation Delivery with Menu Provisioning

The agent fetches the Turtle representation. The response headers MUST restore the “Broken Chain” by providing the identity anchor and the alternate menu.

```
curl -I "https://marineinfo.org/id/institute/36.ttl" -H "Accept: text/turtle"
```

```
HTTP/1.1 200 OK
```

```
Content-Type: text/turtle
```

```
Link: <https://marineinfo.org/id/institute/36-ls.json>; rel="linkset"; type="application/linkset"
```

Menu Delivery (Linkset retrieval)

The agent can additionally check for the availability of other variants through

```
curl -I "https://marineinfo.org/id/institute/36-ls.json" -H "Accept: application/linkset+json"
```

```
HTTP/1.1 200 OK
```

```
Content-Type: application/linkset+json
```

```
{ "linkset":
  [
    { "anchor"   : "https://marineinfo.org/id/institute/36",
      "alternate": [
        { "href"   : "https://marineinfo.org/id/institute/36.ttl",
          "type"    : "text/turtle; charset=utf-8"
        },
        { "href"   : "https://marineinfo.org/id/institute/36.jsonld",
          "type"    : "application/ld+json"
        },
        { "href"   : "https://marineinfo.org/id/institute/36.html",
          "type"    : "text/html; charset=utf-8"
        }
      ]
    },
    { "anchor" : "https://marineinfo.org/id/institute/36.ttl",
      "self"   : [ { "href": "https://marineinfo.org/id/institute/36" } ]
    },
    { "anchor" : "https://marineinfo.org/id/institute/36.jsonld",
      "self"   : [ { "href": "https://marineinfo.org/id/institute/36" } ]
    },
    { "anchor" : "https://marineinfo.org/id/institute/36.html",
      "self"   : [ { "href": "https://marineinfo.org/id/institute/36" } ]
    }
  ]
}
```